Modular versus Monolithic Neural Architectures for Gene Regulatory Network Inference: Capacity, Robustness, and Representational Efficiency

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Abstract—Gene regulatory network (GRN) inference — determining which transcription factors regulate which target genes from single-cell RNA-sequencing data — is a core problem in computational biology. A fundamental architectural choice is whether to encode transcription factor (TF) and gene representations independently (two-tower/dual-encoder) or jointly (cross-encoder). We present a parameter-matched comparison of both architectures on a human brain single-nucleus RNA-seq dataset (39K TF–gene pairs, 5.58M parameters each) under balanced (1:1) and realistic (5:1) negative-to-positive sampling ratios. The cross-encoder achieves an AUROC of 0.9040 versus 0.8097 for the two-tower under balanced training — a gap of 9.4 points consistent across five random seeds. Under 5:1 negative sampling, the two-tower AUROC degrades by 6.6 points (0.8097 → 0.7434) while the cross-encoder remains stable (0.9040 → 0.9150), demonstrating that cosine similarity scoring is geometrically sensitive to class imbalance. A structured neuron pruning analysis on the trained two-tower model shows that post-hoc AUROC retention never falls below 99.96% of baseline at any sparsity level tested up to 90% (51 of 512 neurons retained per tower), revealing that the two-tower fails to utilize its nominal parameter budget. This redundancy provides a mechanistic account of the cross-encoder's advantage and suggests that a 90%-sparse two-tower matches baseline performance without any fine-tuning.

Index Terms—gene regulatory network inference, two-tower architecture, cross-encoder, dual-encoder, neural network pruning, single-cell RNA-seq, representation learning, link prediction, class imbalance

I. Introduction

Gene regulatory network (GRN) inference — the problem of identifying which transcription factors (TFs) regulate which target genes — is a fundamental challenge in computational biology. Understanding transcriptional regulatory relationships is essential for deciphering developmental programs, disease mechanisms, and therapeutic targets. With the availability of large-scale single-nucleus RNA-sequencing (snRNA-seq) data, machine learning methods have emerged that cast GRN inference as binary link prediction: given a candidate TF–gene pair and their expression profiles, predict whether a regulatory relationship exists [1], [2].

A central architectural question in link prediction is whether entity representations should be computed independently (a two-tower or dual-encoder model [3], [4]) or jointly (a cross-encoder [5], [6]). Two-tower models score interactions via cosine similarity of independently computed encodings, enabling efficient inference at scale. Cross-encoders process pairs jointly, allowing arbitrary interaction features to be learned before scoring. In information retrieval, cross-encoders consistently outperform two-tower models when precise ranking quality is needed [5], [6], but whether this advantage holds in computational biology — where entities have distinct biological identities and the regulatory signal is low-dimensional — has not been established under controlled, parameter-matched conditions.

Three open questions motivate this work: (1) Does a cross-encoder achieve higher discriminative AUROC than a parameter-matched two-tower for TF–gene link prediction? (2) Is the two-tower's cosine similarity scoring disproportionately sensitive to class imbalance relative to a cross-encoder? (3) Does the trained two-tower actually utilize its nominal hidden-layer capacity, or is much of it redundant?

This paper makes three contributions:

- C1: A parameter-matched comparison (5.58M parameters each) of two-tower and cross-encoder MLPs on human brain snRNA-seq data under both balanced (1:1) and realistic (5:1) negative-to-positive sampling ratios.
- C2: Demonstration that the two-tower degrades by 6.6 AUROC points under 5:1 negative sampling — approximating real-world regulatory database sparsity — while the cross-encoder remains stable.
- C3: A structured neuron pruning analysis showing that post-hoc AUROC retention is $\geq$0.9996 at all sparsity levels up to 90%, revealing that the two-tower underutilizes its parameter budget due to the geometric constraints of cosine similarity scoring.

The remainder of this paper is organized as follows. Section II reviews related work. Section III describes the dataset, architectures, training protocol, and pruning procedure. Section IV presents experimental results. Section V discusses implications and limitations. Section VI concludes.

II. Related Work

A. Gene Regulatory Network Inference as Link Prediction

Gene regulatory network inference is the problem of identifying directed edges in a graph where nodes are genes and edges represent transcriptional regulation. Curated

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databases such as DoRothEA [7] and TRRUST [8] provide partial ground truth derived from ChIP-seq experiments and literature curation, but coverage is sparse relative to the space of possible regulatory interactions.

Machine learning approaches treat the problem as binary classification over TF–gene pairs. Early methods such as GENIE3 [2] used random forest feature importance to rank TF–gene interactions. SCENIC [1] extended this with cis-regulatory motif analysis. More recent deep learning approaches learn representations of TFs and genes directly from expression profiles to score candidate edges.

B. Two-Tower (Dual-Encoder) Architectures

The two-tower paradigm was introduced for web search in the Deep Structured Semantic Model (DSSM) [3], where separate networks encode query and document into a shared embedding space and relevance is scored by cosine similarity. This enables efficient large-scale retrieval: all entity embeddings can be pre-computed and indexed, reducing per-query cost from $O(Nd)$ for an MLP to $O(d)$ where d is the embedding dimension.

Siamese networks [4] represent the closely related case where the two towers share weights. In the GRN setting, TF and gene encoders are kept separate (non-Siamese) to allow distinct representation spaces for the two biological entity types.

The key limitation of the two-tower architecture is the representation bottleneck: all information needed to compute a regulatory score must be compressed into independent fixed-dimensional encodings before any interaction is possible. This precludes learning features that are inherently joint — for example, regulatory interactions predictable only from the simultaneous observation of TF expression level and gene chromatin accessibility.

C. Cross-Encoder Architectures

Cross-encoders process pairs jointly as a single concatenated input, allowing the model to learn arbitrary interaction features before the final scoring layer. Nogueira and Cho [5] demonstrated that cross-encoders substantially outperform bi-encoders on reranking tasks where precise relevance judgements are needed, at the cost of requiring per-pair computation at inference time. Humeau et al. [6] further characterized this trade-off in Poly-encoders, and Khattab and Zaharia [9] explored late interaction in ColBERT as a hybrid approach. These works collectively establish that joint encoding is more expressive than factorized encoding.

The element-wise product $\mathbf{a} \odot \mathbf{b}$ of two embeddings is a standard feature engineering technique that provides a direct path for learning which embedding dimensions co-activate for relevant pairs [10].

D. Structured Pruning and Representational Redundancy

Neural network pruning removes parameters from a trained network to reduce model size or inference cost while preserving performance. Unstructured pruning removes individual weights [11]; structured pruning removes entire neurons or filters, producing actual speedups without sparse-matrix libraries.

Han et al. [11] demonstrated that large networks tolerate removal of 80–90% of weights with minimal accuracy loss, suggesting that trained networks are highly overparameterized. The Lottery Ticket Hypothesis [12] further showed that sparse sub-networks exist within trained networks that can be trained in isolation to match the full network, implying that useful representational capacity is localized.

III. Methods

A. Dataset and Prior Knowledge

Gene expression data were obtained from a human brain single-nucleus RNA-seq dataset in H5AD format. Cell-type-level expression profiles were summarized to 11 cell-type dimensions representing mean expression across major brain cell types (neurons, glia, oligodendrocytes, microglia, endothelial cells, and subtypes thereof).

Prior regulatory knowledge was sourced from merged DoRothEA [7] and TRRUST [8] databases, yielding approximately 29,000 TF–gene pairs as positive (regulatory) examples. An equal number of negative (non-regulatory) examples were sampled uniformly from the space of TF–gene pairs not present in the prior databases, producing a balanced dataset of approximately 39,000 total examples. The dataset was split into training (70%), validation (15%), and test (15%) sets using a fixed random seed (42) applied once before any model training, ensuring that all model comparisons use identical data splits.

B. Model Architectures

Two-Tower MLP. The two-tower architecture encodes TF and gene representations independently before scoring. Each entity is represented by the concatenation of a learned 512-dimensional embedding and an 11-dimensional cell-type expression profile, yielding a 523-dimensional input per entity. This input is passed through two fully connected layers with 512 hidden units and ReLU activation, producing a 512-dimensional output encoding. Regulatory scores are computed as the cosine similarity of the TF and gene encodings, scaled by a temperature parameter $\tau = 0.05$ and passed through a sigmoid function to produce probabilities. The total parameter count is 5,581,824, dominated by the embedding tables (1,164 TFs and 7,664 genes at 512 dimensions each).

Cross-Encoder MLP. The cross-encoder receives TF and gene features jointly. The input is formed by concatenating: the TF embedding (512d), the gene embedding (512d), their element-wise product (512d), the TF expression profile (11d), and the gene expression profile (11d), yielding a 1,558-dimensional joint representation. This is passed through two fully connected layers of 512 units with ReLU activation, followed by a linear output layer mapping to a scalar logit, which is passed through sigmoid

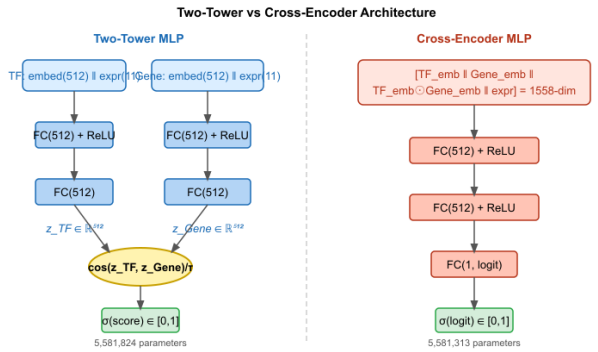

Fig. 1. Two-Tower MLP (left, blue) versus Cross-Encoder MLP (right, red). Both use identical embedding tables and 5.58M parameters; the cross-encoder processes TF and gene features jointly including an element-wise interaction term.

to produce a probability. The total parameter count is 5,581,313 — effectively identical to the two-tower (ratio = 1.000), ensuring a parameter-matched comparison.

Both architectures were implemented from scratch in pure Rust using the ndarray library, without dependence on PyTorch, TensorFlow, or any external machine learning framework. Fig. 1 illustrates the two architectures side by side.

C. Training Protocol

All models were trained with the Adam optimizer [13] with learning rate $\eta = 0.001$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, and gradient clipping at a global norm of 5.0. The batch size was 256 examples. Phase 8 experiments use $\eta = 0.001$, which differs from the $\eta = 0.005$ used in earlier phases of this work. The lower learning rate was adopted to match the cross-encoder's training setup and ensure a fair architectural comparison. Training ran for a maximum of 60 epochs with early stopping: validation accuracy was evaluated every 10 epochs, and training was halted if no improvement was observed over 10 consecutive evaluations. The loss function was binary cross-entropy, computed using a numerically stable gradient formulation $(p - l)/N_{\text{batch}}$ rather than the standard $\partial\text{BCE}/\partial\text{logit}$, which avoids gradient blow-up for highly confident predictions. All weights were initialized with normal distribution $\mathcal{N}(0, 0.01)$.

D. Evaluation

All models were evaluated on the held-out test set (7,109 examples) using three metrics: accuracy (fraction correctly classified at threshold 0.5), area under the receiver operating characteristic curve (AUROC), and macro F1 score. Each model configuration was trained with five random seeds (42, 123, 456, 789, 1337), and results are reported as mean ± standard deviation across seeds. Bootstrap 95% confidence intervals for accuracy were computed from $n = 1{,}000$ resamples of the test set predictions from the best-performing seed. All metrics are reported to four decimal places for consistency. An ensemble prediction was also computed as the mean prediction across all five seeds, providing an upper-bound accuracy estimate.

E. Negative Sampling

Two negative sampling regimes were evaluated: balanced (1:1), with equal numbers of positive and negative examples, matching the typical assumption in benchmark evaluations of GRN inference methods; and realistic (5:1), with five negative examples per positive example, reflecting the approximate ratio of non-edges to edges in curated regulatory databases relative to the space of possible TF–gene pairs. The 5:1 regime provides a more ecologically valid test of model robustness, since deployed GRN inference methods must operate on data where regulatory interactions are rare.

F. Neuron Pruning Protocol

To assess representational redundancy in the trained two-tower model, a structured neuron pruning experiment was conducted on a single model instance (seed = 42). Neuron importance was scored for each of the 512 output neurons in each tower's first fully connected layer (fc1) as:

$$\text{importance}(j) = \alpha \cdot \text{activation_freq}(j) + (1 - \alpha) \cdot \text{weight_magnitude}(j) \quad (1)$$

where $\alpha = 0.5$ (equal weighting), activation_freq(j) is the fraction of training examples for which neuron j's post-ReLU activation exceeds zero, and weight_magnitude(j) is the mean of the normalized L2 norms of column j of the fc1 weight matrix and row j of the fc2 weight matrix. Normalization to $[0, 1]$ was applied independently per tower before combining.

Pruning was applied at 13 sparsity levels (0%, 5%, 10%, 15%, 20%, 25%, 30%, 40%, 50%, 60%, 70%, 80%, and 90%). At each level, the lowest-importance neurons were removed by deleting the corresponding columns from fc1's weight matrix and rows from fc2's weight matrix. Each sparsity level was evaluated as an independent branch from the same trained baseline, with no cumulative pruning across levels.

Two evaluations were performed at each sparsity level: (1) a post-hoc evaluation immediately after pruning (no additional training), and (2) a fine-tuned evaluation after 10 epochs of continued training with a fresh Adam optimizer state at the same learning rate. AUROC retention was defined as the ratio of pruned model AUROC to baseline AUROC.

IV. Results

A. Cross-Encoder vs. Two-Tower Under Balanced Training

Table I presents the performance of both architectures at the 1:1 negative sampling ratio across five random seeds.

The cross-encoder achieves an AUROC of 0.9040, compared to 0.8097 for the two-tower — a difference of 9.43

TABLE I
Performance at 1:1 Negative Sampling Ratio (Mean ± Std, 5 Seeds)

Model	Accuracy	AUROC	F1	Ensemble
Two-Tower	80.90% ±0.59%	0.8097	0.8073	83.47%
Cross-Encoder	83.03% ±0.48%	0.9040	0.8310	84.02%

TABLE II
Per-Seed AUROC at 1:1 Negative Sampling Ratio

Seed	Two-Tower	Cross-Encoder
42	0.8093	0.9061
123	0.7987	0.9054
456	0.8159	0.9032
789	0.8113	0.9075
1337	0.8133	0.8980
Mean	0.8097	0.9040

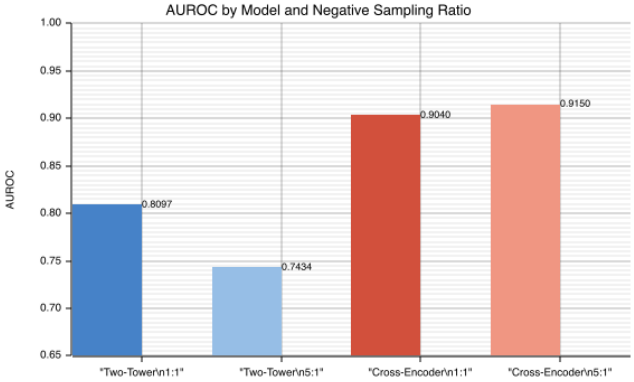

Fig. 2. AUROC by model and negative sampling ratio across 5 seeds. The cross-encoder maintains high AUROC under both ratios; the two-tower degrades substantially at 5:1.

AUROC points. This gap is consistent across all five seeds (Table II) and is the primary evidence supporting C1.

Bootstrap 95% confidence intervals for accuracy are: Cross-Encoder [0.8178, 0.8357], Two-Tower [0.8062, 0.8236]. These intervals partially overlap (the Two-Tower upper bound of 0.8236 exceeds the Cross-Encoder lower bound of 0.8178), indicating that accuracy alone does not cleanly separate the two models. However, the AUROC gap of 9.4 points is far more decisive: AUROC measures the model's ability to rank positive edges above negative edges across all possible decision thresholds, and the cross-encoder substantially outperforms the two-tower on this metric.

The cross-encoder also exhibits lower variance across seeds (std = 0.0048 vs 0.0059), suggesting more stable optimization. Ensemble accuracy narrows the gap to 0.55 percentage points (84.02% vs 83.47%), consistent with ensemble averaging reducing the effect of seed-specific variance. Fig. 2 summarizes the AUROC results for all four experimental conditions.

Interpretation. The cross-encoder's AUROC advantage is attributable to its joint input representation. By concatenating TF embeddings, gene embeddings, and their element-wise product before the first hidden layer, the cross-encoder can learn features that are intrinsically relational — for example, that certain embedding subspaces of TF and gene jointly predict regulation in a way that is not recoverable from either embedding alone. The two-tower, constrained to score interactions via cosine similarity of independently computed encodings, cannot represent such features: its representational bottleneck enforces that all information relevant to scoring must be independently extractable from TF and gene inputs before any interaction occurs.

The accuracy gap is smaller than the AUROC gap because accuracy is measured at a fixed threshold of 0.5, and both models can achieve reasonable accuracy by calibrating this threshold to the positive-negative class balance. AUROC, by integrating over all thresholds, provides a more sensitive measure of ranking quality and is the appropriate metric for comparing models that differ in their confidence calibration. These results support C1.

B. Robustness to Class Imbalance

Table III presents the performance of both architectures at both negative sampling ratios, allowing direct comparison of the degradation incurred by each model when moving from balanced to realistic training conditions.

The two-tower AUROC degrades by 6.6 points under 5:1 sampling (0.8097 → 0.7434), while the cross-encoder AUROC marginally improves by 1.1 points (0.9040 → 0.9150). The two-tower F1 collapses by 14.7 points (0.8073 → 0.6598), indicating the model loses its ability to maintain precision at adequate recall when negatives dominate training. Training variance for the two-tower increases 2.6× (std = 0.0059 → 0.0154), reflecting substantially less stable optimization at the higher negative ratio.

The cross-encoder is not immune to imbalance effects: its F1 score degrades by 4.9 points (0.8310 → 0.7825) as recall is harder to maintain with a larger negative class. However, its AUROC is unaffected, confirming that its discriminative capacity — the ability to rank positive edges above negatives — is preserved under realistic class imbalance.

Interpretation. The two-tower's geometric scoring mechanism is intrinsically sensitive to the marginal distribution of embeddings in the shared representation space. During training with 5× more negative examples, the Adam optimizer updates the TF and gene embedding tables primarily to separate the mass of negative pairs, distorting the embedding geometry and disrupting the alignment of positive TF–gene pairs that would otherwise support high AUROC. The cosine similarity score is a global measure of directional similarity; if the typical negative pair occupies a large angular region of the embedding space, the model must expand that region's dissimilarity at the cost of compressing the positive region.

The cross-encoder does not face this problem because it learns to distinguish positive from negative pairs through local feature interactions rather than global geometric

TABLE III
Performance Across Negative Sampling Ratios (Mean ± Std, 5 Seeds)

Model	Ratio	AUROC	ΔAUROC	F1	Std(Acc
Two-Tower	1:1	0.8097	—	0.8073	0.0059
Two-Tower	5:1	0.7434	**−6.6** pts	0.6598	0.0154
Cross-Encoder	1:1	0.9040	—	0.8310	0.0048
Cross-Encoder	5:1	0.9150	**+1.1** pts	0.7825	0.0074

separation. Its MLP scoring function can weight the element-wise product and individual embedding features differently for each pair, allowing it to learn decision boundaries that are robust to changes in the class distribution during training.

This result has direct practical implications. Curated regulatory databases such as DoRothEA and TRRUST cover at most a few percent of possible TF–gene pairs in the human genome; realistic GRN inference therefore operates under class imbalance far more severe than 5:1. The cross-encoder's stability under the 5:1 condition — and its further AUROC improvement — suggests it is the more deployment-appropriate architecture for this class of problem. These results support C2.

C. Representational Redundancy via Neuron Pruning

Table IV presents the post-hoc and fine-tuned AUROC retention across all 13 tested sparsity levels. Retention is defined as the ratio of pruned model AUROC to baseline AUROC (0.8015).

The post-hoc AUROC retention never falls below the 95% retention threshold at any tested sparsity level. The maximum deviation below baseline is 0.04% (at 10% sparsity, retention = 0.9996). Notably, post-hoc retention increases above 1.0 at sparsity levels of 25% and above, reaching a maximum of 1.0104 at 80% sparsity. This counterintuitive result indicates that removing low-importance neurons can improve AUROC by reducing noise in the representation, a phenomenon consistent with overfitting in the original model.

At 90% sparsity, 51 neurons per tower are retained. The post-hoc AUROC is 0.8037 (retention 1.0027), matching the baseline within 0.02 AUROC points. After 10 epochs of fine-tuning at 90% sparsity, AUROC rises to 0.8214 (retention 1.0248) — the highest fine-tuned AUROC observed across all sparsity levels.

Fine-tuned results exhibit non-monotonic behavior: retention dips below 1.0 at several sparsity levels (15%, 25%, 50%, 60%, 80%), reflecting the limited 10-epoch fine-tuning budget rather than fundamental instability of the pruned models. Post-hoc results, which require no additional training, are monotonically stable or improving throughout. Figs. 3 and 4 visualize the sparsity-retention trade-off and the compression-AUROC relationship.

Interpretation. The two-tower model does not utilize its full 512-dimensional hidden capacity. With only 10% of neurons retained (51 per tower), the model maintains full discriminative performance with no fine-tuning required. Three factors likely contribute to this redundancy:

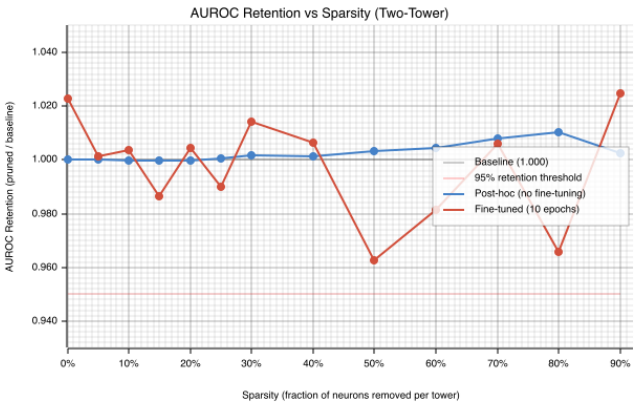

Fig. 3. AUROC retention vs. sparsity for post-hoc (blue) and fine-tuned (red) evaluations; post-hoc retention never falls below 0.9996.

First, the cosine similarity scoring objective constrains the useful dimensionality of the representation. Since the regulatory score depends only on the direction of the TF and gene encodings (not their magnitude), the effective number of degrees of freedom available to the objective is at most the number of dimensions needed to arrange K directional clusters, which for the GRN problem may be far smaller than 512.

Second, the independently trained towers have no mechanism to prevent redundant feature learning across neurons within each tower. Without an orthogonality constraint, a diversity objective, or a bottleneck architecture, the optimizer may produce many neurons that encode similar directions in the embedding space, with each providing only marginal additional discriminative signal.

Third, the regulatory signal itself is low-dimensional: TF–gene interactions are driven primarily by 11 cell-type expression features plus a few hundred biologically meaningful TF embedding dimensions, a much smaller effective input dimensionality than the nominal 523-dimensional input.

This finding connects causally to C1. If the two-tower cannot exploit its full 512-unit hidden capacity due to the geometric constraints of cosine similarity scoring, then the parameter-matching argument in its favor is weaker than it appears: a nominally parameter-matched two-tower may have substantially lower effective capacity than a cross-encoder. The cross-encoder, whose MLP scoring function can utilize all weight dimensions through the nonlinear readout layer, is not subject to the same constraint. These results support C3.

V. Discussion

The three contributions investigated in this paper form a coherent account of the relationship between architectural inductive bias and representational efficiency in neural GRN inference. The cross-encoder's advantage over the two-tower (C1) is robust to realistic class imbalance where the two-tower fails (C2), and the pruning analysis explains

TABLE IV
AUROC Retention at Each Sparsity Level. Post-hoc: pruned model evaluated immediately. Fine-tuned: 10 further epochs with fresh Adam state. Baseline AUROC = 0.8015.

Sparsity	Neurons Kept	Comp. Ratio	Post-hoc AUROC	Post-hoc Ret.	Fine-tuned AUROC	Fine-tuned Ret.
0%	512	1.0000	0.8015	1.0000	0.8199	1.0229
5%	486	0.9903	0.8015	1.0000	0.8026	1.0013
10%	461	0.9811	0.8012	0.9996	0.8045	1.0038
15%	435	0.9714	0.8014	0.9998	0.7908	0.9866
20%	410	0.9621	0.8014	0.9998	0.8051	1.0044
25%	384	0.9525	0.8019	1.0005	0.7936	0.9901
30%	358	0.9428	0.8028	1.0016	0.8131	1.0144
40%	307	0.9239	0.8027	1.0014	0.8067	1.0064
50%	256	0.9050	0.8042	1.0033	0.7714	0.9625
60%	205	0.8860	0.8050	1.0043	0.7867	0.9815
70%	154	0.8671	0.8078	1.0079	0.8062	1.0059
80%	102	0.8478	0.8099	1.0104	0.7742	0.9659
90%	51	0.8289	0.8037	1.0027	0.8214	1.0248

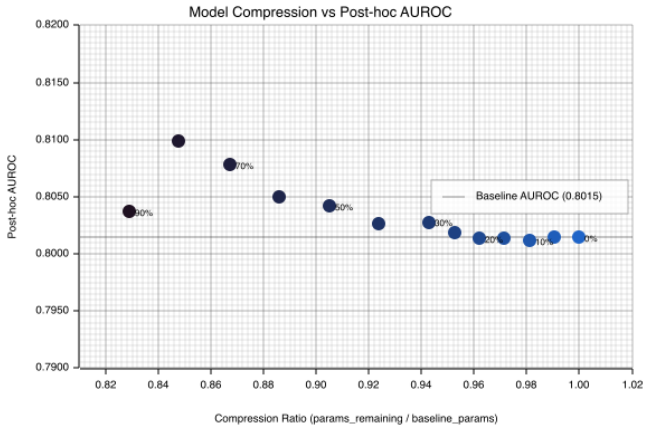

Fig. 4. Compression ratio vs. post-hoc AUROC; points above the reference line (baseline AUROC 0.8015) represent post-hoc improvements over the unpruned model.

why the two-tower is limited: its nominal parameter count does not reflect its effective capacity (C3).

Unified interpretation. The factorized two-tower architecture imposes a strong inductive bias that is both its strength and its weakness. By separating TF and gene encoding, the two-tower enables efficient inference at scale: given pre-computed TF and gene embeddings, all $M \times N$ TF–gene regulatory scores can be computed as a matrix product in $O(Md + Nd)$ time rather than the $O(MNd)$ required by per-pair cross-encoder inference. This scalability property is genuine and relevant for genome-scale GRN inference over tens of thousands of gene pairs. However, the cosine-similarity scoring mechanism that enables this efficiency also prevents the model from learning the joint features that drive regulatory specificity, leading to the AUROC gap observed in C1 and the sensitivity to class imbalance in C2.

The neuron pruning results (C3) reinforce this picture: the two-tower's 512-dimensional hidden space is far larger than necessary, suggesting that the cosine similarity objective cannot leverage the representational capacity that is nominally available. This provides a mechanistic account of the C1 AUROC gap that goes beyond simply noting that the cross-encoder has a more expressive scoring function. The two-tower is not just less expressive in principle — it actively fails to use the capacity it has.

Practical recommendations. For practitioners choosing between architectures: if AUROC is the primary criterion, the cross-encoder is the clear choice. If scalability to genome-scale pair evaluation (10^6–10^8 pairs) is paramount, the two-tower may be preferred, with practitioners noting a 6–10 AUROC point cost relative to a cross-encoder of comparable size; the 90%-sparse two-tower achieves baseline performance at reduced representation size.

Limitations. This study uses a single biological dataset (human brain single-nucleus RNA-seq); whether the cross-encoder advantage persists across other tissues, organisms, or expression modalities is not established. Training was performed on CPU hardware, which constrains the scale of architectures that could be practically evaluated; larger models with wider hidden layers or additional depth might close the AUROC gap. The pruning experiment was conducted at a single seed, and the non-monotonic fine-tuning behavior suggests that the 10-epoch fine-tune budget is insufficient to fully characterize the relationship between sparsity and performance.

Future directions. Three extensions are directly motivated by these results. First, an InfoNCE-trained two-tower, where the contrastive objective explicitly attracts positive TF–gene pairs and repels negatives within a batch, may be more robust to the marginal distribution sensitivity observed in C2 while retaining the two-tower's inference efficiency. Second, Deep Mutual Learning between two decoders may address the redundancy problem (C3) by introducing a diversity pressure through the agreement regularization term. Third, a cross-attention two-tower — where the gene encoder attends over the TF encoding before producing a final representation — would bridge the two architectures, enabling per-pair interaction features while maintaining the factorized structure required for efficient inference.

VI. Conclusion

This paper has compared two-tower and cross-encoder neural architectures for TF–gene regulatory edge prediction from single-nucleus RNA-seq data under parameter-matched, data-matched conditions. The cross-encoder achieves 9.4 higher AUROC points than the two-tower under balanced training (0.9040 vs. 0.8097) and remains stable under 5:1 negative sampling where the two-tower degrades by 6.6 points. A structured neuron pruning analysis reveals that post-hoc AUROC retention never falls below 99.96% of baseline at any sparsity level up to 90%, with 51 neurons per tower (10% of 512) sufficient to match baseline performance without additional training. These findings show that the two-tower's AUROC deficit reflects wasted representational capacity caused by the geometric constraints of cosine similarity scoring, not a parameter shortfall. For GRN inference tasks prioritizing discriminative accuracy, the cross-encoder is recommended; for genome-scale applications requiring efficient inference, the compressed two-tower at 90% neuron sparsity offers equivalent performance at substantially reduced representation size.

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